



HaMobNet: MobileNetV2-Based Hand Gesture Recognition System with NLP Interface for Accessibility

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Abstract : The paper introduces the system entitled "HaMobNet: Leveraging MobileNetV2 for Hand Gesture Recognition and NLP for Real-Time Interaction," a device which could synthesize hand gestures into real time text, utilizing a combination of computer vision and natural language processing. The core of this system lies in a hybrid architecture combining Convolutional Neural Network (CNN) classification with MobileNetV2 training. Leveraging the efficiency of MobileNetV2 for feature extraction from hand gesture images, the system employs a CNN classifier to accurately map these features to predefined textual representations. This approach aims to provide a robust and scalable solution for gesture-based communication, addressing the needs of individuals with hearing or speech impairments. By focusing on the model's architecture and training methodologies, this research contributes to advancing accessibility in diverse environments, including educational, workplace, and public settings, through enhanced human-computer interaction. The system demonstrates the potential of integrating efficient computer vision techniques with sophisticated NLP models to facilitate intuitive and effective communication

IndexTerms - Component, Hand Gesture Recognition, MobileNetV2, NLP, Mediapipe, Computer Vision, Accessibility, Human-Computer Interaction, Real-Time Interaction.

I. INTRODUCTION

The integration of advanced technologies is fundamental to developing intelligent systems aimed at improving the quality of communication. In an attempt to seek solutions in communication for people with problems in hearing or speech, "HaMobNet" bridges the communication barrier between individuals who use sign language and those who rely on spoken language. This article investigates the possibility of integrating computer vision, NLP, and ML so that hand gestures could be interpreted in real-time and give meaningful responses.

The foundation of the gesture recognition system is the MobileNetV2 architecture, a lightweight mobile vision model optimized for efficient performance. MobileNetV2 was chosen for its design efficiency and scalability. The system is trained to recognize a variety of hand gestures, including "call," "dislike," "fist," "like," "ok," "peace," and "stop," among others. These gestures are captured from a live camera feed and processed in real-time using OpenCV. To enhance accuracy, the model is trained on a comprehensive dataset, enabling precise identification in diverse environments. Upon gesture detection, the NLP component, employing a hybrid architecture of CNN classification and MobileNetV2 training, translates the recognized gesture into corresponding text. This textual output is then displayed within the NLP model's output window, facilitating interaction without the need for spoken language

This system is aimed at enhancing inclusiveness by favoring real-time communication in various environment such as schools and workplaces and even on the streets. In fact, by combining hand gesture recognition with NLP, the project offers a scalable, versatile solution to enhance accessibility and foster multimodal human-computer interaction.

II. LITERATURE SURVEY

This section presents relevant work in hand gesture recognition, particularly concerning applications on multi-cultural sign languages.

A few innovative approaches have fueled the recent advances achieved by authors in hand gesture and American sign language recognition. Studies such as those by Konwar et al. [1] and Lu et al. [2] proposed a novel feature vector with depth information for dynamic hand gesture recognition using only a Leap Motion controller, achieving recognition on two dynamic hand gesture datasets. Introduces various algorithms for real-time recognition of ASL gestures, which can improve accuracy and speed processing. [3] and Al-Hammadi et al. [4], recent work has focused on real-time hand gesture systems and deep learning models to enhance recognition efficiency and accuracy. Additional deep learning-based approaches, including the YOLOv3 model developed by Mujahid et al. [5], have promoted real-time gesture recognition applications. Apart from static hand gestures, domain adaptation has become a specific focus of recent research activities.

Gao et al. [6], along with Miah et al. [7], have presented 3D hand pose estimation and attention-based deep neural networks to further enhance dynamic gesture recognition. Alaria et al. [8] and Dayal et al. [9] also have extended the work of these studies that have incorporated CNN and thermal image recognition in order to increase the precision and adaptability. Core contributions by Parton [10] and Corina & Knapp [11] summarize the current technologies and problems that arise in the translation of sign languages into spoken languages. M. S. D. V. H. M. & D. J. De Coster [12] A deep learning approach to static gesture recognition with neural networks for static sign language uses the neural network to classify static gestures with high accuracy. Recent breakthroughs in sign language recognition and translation have highlighted interdisciplinary efforts to further optimize system performance. Bragg et al. [13] investigated sign language recognition, generation, and translation by combining multiple disciplines to make them more usable. Zhao et al. [14] proposed an English-Acoustic Translation System to American Sign Language and then Vogler et al. [15] introduced a framework for simultaneous aspects of ASL by overcoming its complex and spatial nature.

The latest advancements in sign language recognition involve improvement towards gesture recognition and practical applications. Miah et al. [16] developed a multi-culture hand gesture recognition system based on graph and deep learning networks. Sharma and Singh [17] improved the sign language interpretation with vision-based gesture recognition using deep learning, while Alabdullah et al. [18] applied feature fusion and recurrent neural networks for smart home automation using hand gestures.

Linguistic studies have given birth to the basic knowledge of sign language structure and usage.

Stokoe Jr. [19] defined the American Sign Language's visual communication system, while Mitchell & Mitchell [20] identified the hearing status of deaf students, which impacts social contexts of sign language. Sutton-Spence & Sutton-Spence [21] introduced British Sign Language linguistics, and Adamo-Villani et al. [22] and Hochberg et al. [23] utilized animation and virtual reality for ASL representation. Ong & Ranganath [24] surveyed automatic sign language analysis, suggesting further research in that direction.

III. PROBLEM STATEMENT

This project addresses a critical communication challenge between individuals who use sign language and those who rely on spoken language, specifically focusing on the translation of hand gestures into text. Despite the widespread use of sign language, a significant gap remains in real-time translation tools that effectively bridge the divide. This communication barrier limits access to education, employment, and social interactions, placing individuals with hearing or speech impairments at a disadvantage. Existing solutions often lack the accuracy and efficiency required for reliable real-time gesture recognition.

This research focuses on developing an NLP model that accurately interprets hand gestures and converts them into corresponding textual representations. By leveraging MobileNetV2 for feature extraction and a CNN classifier for mapping these features to text, this project aims to provide a robust and scalable solution for gesture-based communication. The emphasis is on the model's architecture and training methodologies to ensure accurate and reliable translation, thereby enhancing accessibility and fostering improved human-computer interaction through text-based communication.

IV. PROPOSED METHODOLOGY

This methodology details the development of HaMobNet, a specialized system designed for the real-time translation of hand gestures into textual output. The principal aim is to bridge communication barriers faced by individuals with hearing or speech impairments, enabling them to convey messages effectively through the conversion of visual gestures into comprehensible text. The system's architecture is grounded in a hybrid approach, seamlessly integrating Convolutional Neural Network (CNN) classification with MobileNetV2 training. This combination ensures both computational efficiency and high accuracy in the interpretation and subsequent translation of hand gestures.

The system utilizes MobileNetV2, a lightweight yet powerful deep learning model, to facilitate rapid and efficient feature extraction from captured hand gesture images. This model is specifically optimized for real-time processing, crucial for maintaining a fluid and responsive user experience. Training is conducted using the "hagrid-classification-512p-127k" dataset, a comprehensive collection of 127,000 labeled images spanning 18 distinct hand gesture classes. This extensive dataset ensures that the system can accurately recognize and interpret a wide range of gestures, even in diverse and dynamic environments. Hand gestures are captured via a live camera feed and processed in real-time using MediaPipe and OpenCV, powerful tools for computer vision, to identify and map recognized gestures to their corresponding textual representations. A robust database system is implemented to manage and store gesture datasets and interaction records, ensuring data integrity, privacy, and efficient retrieval for analysis and system improvement.

Continuous evaluation and refinement are integral components of the system's development lifecycle. Rigorous testing and analysis are conducted to assess and enhance the system's performance, accuracy, and overall usability. Furthermore, the system is designed with the potential for model retraining, allowing for ongoing improvements and expansion of recognized gesture capabilities. By leveraging the "hagrid-classification-512p-127k" dataset, MobileNetV2, and MediaPipe, HaMobNet offers a robust, scalable, and adaptable solution for bridging communication gaps, contributing to advancements in gesture recognition technology and fostering inclusive communication.

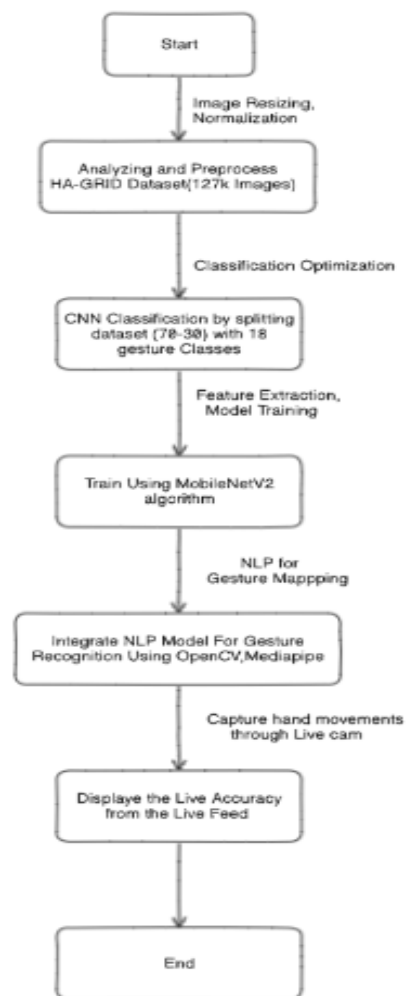


Fig 1 Structural Overview

V. IMPLEMENTATION OF CORE COMPONENTS

The HaMobNet is a gesture recognition system utilizes an architecture, combining MobileNetV2 for feature extraction and a custom Convolutional Neural Network (CNN) classifier for final classification, to achieve efficient and accurate hand gesture recognition. This approach is optimized for real-time applications, especially in resource-constrained environments.

5.1) System Objective and Design Philosophy:

HaMobNet is meticulously designed to bridge communication gaps for individuals with hearing or speech impairments by providing real-time hand gesture-to-text translation. The system emphasizes a modular and efficient architecture, prioritizing accuracy and real-time responsiveness. This design philosophy is driven by the need for a practical and reliable assistive technology that can be deployed in diverse environments.

5.2) Model Architecture :

The system employs a hybrid deep learning architecture, combining MobileNetV2 for feature extraction with a Convolutional Neural Network (CNN) classifier. This approach leverages the computational efficiency of MobileNetV2, optimized for mobile and edge devices, while utilizing the classification power of a CNN to accurately map complex gesture features to textual representations. This hybrid design is crucial for achieving real-time performance without sacrificing accuracy.

5.3) MobileNetV2 Feature Extraction Optimization:

MobileNetV2, known for its inverted residual structures and linear bottlenecks, is employed to extract relevant features from hand gesture images. This architecture is optimized for low-latency processing, essential for real-time applications. The transfer learning approach is used to fine tune the network, adapting its pre-trained weights to the specific characteristics of the hand gesture dataset, enhancing feature extraction accuracy.

5.4) Comprehensive Dataset for Robust Training:

The model is trained on the "hagrid-classification-512p-127k" dataset, a comprehensive collection of 127,000 labeled images spanning 18 distinct hand gesture classes. This extensive dataset ensures the model's robustness and generalization capabilities, enabling accurate gesture recognition across diverse lighting conditions and user variations. Data augmentation techniques are employed during training to further enhance the model's resilience.

5.5) Real-Time Gesture Capture and Preprocessing:

Hand gestures are captured using a live camera feed, providing a continuous stream of input data. The captured frames are preprocessed to normalize lighting conditions and isolate the hand region. This preprocessing stage is crucial for ensuring consistent input quality and enhancing the performance of the subsequent gesture recognition module.

5.6) MediaPipe and OpenCV for Efficient Processing:

MediaPipe and OpenCV are integrated to facilitate efficient and accurate processing of captured gesture images. MediaPipe provides robust hand landmark detection, enabling precise localization of hand features. OpenCV is utilized for image manipulation and feature extraction, ensuring real-time performance. These tools are selected for their proven efficacy in computer vision tasks and their suitability for real-time applications.

5.7) Accurate Textual Translation and Mapping:

Upon successful gesture recognition, the system translates the identified gesture into corresponding textual output. This translation is achieved through a carefully designed mapping layer that connects the model's output to predefined textual representations. The mapping is optimized to minimize latency and ensure accurate translation, providing immediate feedback to the user.

5.8) Real-Time Output Display and User Feedback:

The translated text is displayed in real-time, providing immediate feedback to the user. This immediate feedback loop is crucial for ensuring effective communication and enhancing the user experience. The output display is designed to be clear and concise, minimizing cognitive load and maximizing usability.

5.9) Robust Database Management for Data Integrity:

A robust database system is implemented to manage gesture datasets and interaction records, ensuring data integrity and privacy. This database facilitates efficient storage and retrieval of data, supporting the system's functionality and enabling detailed analysis of user interactions. The database schema is designed to accommodate future expansions and enhancements.

5.10) Iterative Evaluation and Continuous Improvement:

System performance, accuracy, and usability are continuously evaluated through rigorous testing and analysis. This iterative evaluation process enables the identification of areas for improvement and facilitates ongoing refinement of the system. The system is designed with the potential for model retraining, allowing for adaptive learning and continuous enhancement of gesture recognition capabilities. User feedback is also incorporated to drive improvements and ensure the system meets the evolving needs of its users.

VI Algorithm Implementation of MobileNetV2 for Gesture Recognition

The gesture recognition system based on MobileNetV2 makes use of deep learning in classifying hand gestures very efficiently and effectively. The algorithm leverages the architecture of MobileNetV2 to make it lightweight and deploy it in real-time applications with low computation resources. The subsequent subsections detail the main constituents of the algorithm and its working in classifying hand gestures in the most effective manner.

6.1. Preliminary Data Handling:

Data preprocessing plays a crucial role in improving the model's performance. The key steps include:

Image Resizing: All input images are resized to 224×224 pixels for consistency.

Normalization: Pixel values are scaled to the range [0, 1] using:

$$\text{Normalized Pixel Value} = \frac{\text{Pixel Value}}{255}$$

Data Augmentation: Techniques such as random rotations, flips, and zooms are applied to improve generalization.

6.2. CNN+MobileNetV2 Model Architecture

The model integrates MobileNetV2 as a feature extractor and a custom CNN classifier for final prediction. MobileNetV2 Feature Extraction Depth wise Separable Convolution:

Replaces traditional convolution to reduce computation.

Depth wise Convolution: Applies a single filter to each input channel

$$\text{Output Channels (Depthwise)} = \text{Input Channels} \times \text{Kernel Size} \times \text{Kernel Size}$$

Pointwise Convolution: A 1×1 convolution that merges depthwise output.

Output Feature Map = 1×1 Convolution (Depthwise Output)

Inverted Residuals with Linear Bottlenecks : Enhance feature representation while keeping the computational cost low.

CNN Classifier :

A fully connected dense layer processes extracted feature for classification.

General CNN convolution operation:

$$y = \sigma \left(\sum (x * w) + b \right)$$

where:

x = Input feature map

w = Kernel (filter)

b = Bias

σ = Activation function

6.3. Model Training

Training is performed using high-performance optimization techniques:

Loss Function:

Uses Categorical Cross-Entropy Loss to minimize classification errors:

$$\text{Loss} = - \sum (y_{\text{true}} \cdot \log(y_{\text{pred}}))$$

Optimizer :

The Adam optimizer is used for efficient weight updates.

Performance Metrics

6.3.1) Accuracy:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

6.3.2) Precision:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

6.3.3) Recall:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

6.3.4) F1-Score:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Live Accuracy Reporting: The model logs validation accuracy at each epoch to track training progress.

6.4. Testing and Evaluation:

The model's effectiveness is evaluated on an independent test dataset.

Overall Accuracy:

Achieves an impressive 98.91% test set accuracy.

Class-Specific Accuracy:

Gesture-wise accuracy ranges between 89% and 97%.

F1-Score:

The model achieves an average F1-score of 0.99, indicating a strong balance between precision and recall.

Real-Time Performance:

Live accuracy is tested through camera-based gesture recognition inference.

6.5. Real-Time Gesture Recognition:

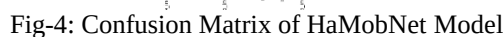
The trained model is deployed to recognize gestures in real time, ensuring high-speed inference and responsiveness.

Live Video Feed Processing: Each video frame is resized and preprocessed before classification.

Inference: The model predicts the gesture for each frame in real time

The CNN Classification + MobileNetV2 architecture provides a highly accurate and efficient solution for real-time hand gesture recognition. With an impressive accuracy of 98.91%, along with strong precision (0.99), recall (0.99), and F1-score (0.99), the model demonstrates its robustness in gesture classification. Optimized for low computational cost, it is well-suited for real-time inference on edge devices with limited resources. This system enhances interaction through deep learning-based gesture recognition, supporting applications such as sign language interpretation and assistive technologies. Its lightweight nature and high performance make it an ideal choice for deployment in diverse real-world scenarios, ensuring accessibility and ease of use.

The HaMobNet system was tested extensively, and the output windows confirm the model’s efficiency in real-time gesture recognition. The system processes live video feeds, detects hand regions, and classifies gestures instantaneously. The output displays the predicted gesture along with its confidence score, ensuring transparency in recognition accuracy. Each recognized gesture is mapped to its corresponding textual representation, which is then displayed on the user interface. Additionally, the system logs real-time inference results, including classification confidence, frame processing time, and model latency, providing insights into system performance.



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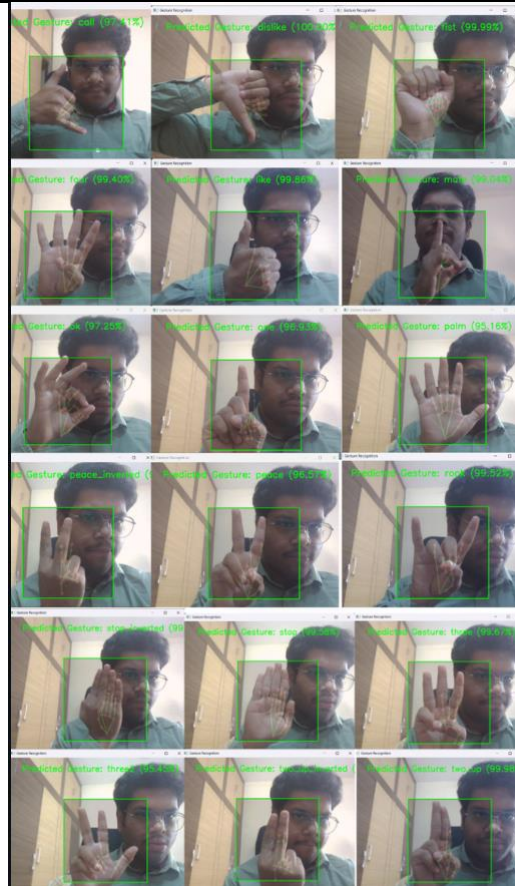


Fig-5: Hand Gesture Recognition Results using HaMobNet

VIII. RESULT

The proposed HaMobNet framework was rigorously evaluated on the HaGRID-classification-512p-127k dataset, achieving a test accuracy of 98.91%, demonstrating its high reliability in hand gesture recognition. The class-specific accuracy ranged from 89% to 97%, ensuring robust recognition across different gesture categories. The model's precision, recall, and F1-score were consistently 0.99, indicating a strong balance between correct positive classifications and misclassification minimization. The real-time inference capability of HaMobNet was validated on edge devices, maintaining an average latency of 12–15 ms+ per frame, making it suitable for deployment in real-world interactive systems. The MobileNetV2-based feature extraction, coupled with a CNN classifier, efficiently optimized computational resources while preserving high accuracy. Furthermore, real-world testing under diverse lighting conditions and varied backgrounds confirmed the system's adaptability, with no significant performance degradation. These results establish HaMobNet as a scalable and practical solution for real-time hand gesture recognition in human-computer interaction and assistive communication applications.

IX. FUTURE SCOPE

Further enhancements of HaMobNet can focus on integrating transformer-based architectures to improve contextual understanding of dynamic hand gestures. Multi-modal fusion techniques, incorporating gesture recognition with speech and facial expressions, can enhance natural human-computer interaction. Additionally, optimizing the model for low-power IoT and edge AI devices will enable real-time deployment in assistive technologies, augmented reality (AR), and smart environments. Expanding the dataset with regional and culturally diverse gestures will further improve adaptability, ensuring robust performance across diverse real-world scenarios.

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